

Wildfire Prediction Case Study – Student Rubric

DS 4002: Spring 2026

Submission format: GitHub repository (submitted by link to Canvas)

General Description: Using the provided California weather dataset (1984–2025), build and evaluate two machine learning models that predict whether a wildfire will occur on a given day. Submit a GitHub repository containing your code, outputs, and a written report.

Why am I doing this? Wildfires are one of the most destructive and increasingly frequent natural disasters in the United States. The 2025 LA and San Diego fires resulted in over 400 deaths and tens of thousands of displaced residents. This case study gives you hands-on experience with a real-world binary classification problem where the cost of errors is not symmetric — missing a fire day is far worse than a false alarm.

- Course Learning Objective: apply machine learning methods to a real-world time series prediction problem and communicate findings clearly.

What am I going to do? Using the provided data and starter notebooks, you will:

- Explore and visualize the dataset to identify seasonal patterns related to wildfire occurrence
- Build a logistic regression model and a random forest model to predict FIRE_START_DAY
- Evaluate both models using accuracy, precision, recall, F1-score, and ROC-AUC
- Write a short report recommending which model an emergency agency should deploy and why

How will I know I have succeeded? You will meet expectations on this case study when you follow the criteria in the rubric below.

Spec Category	Spec Details
Formatting	• One GitHub repository submitted via link on Canvas • The top level of the repository should contain: • A README.md file • A LICENSE.md file (MIT license) • A SCRIPTS folder • A DATA folder • An OUTPUT folder
README.md	• Goal: Orient anyone who visits your repository so they can get their bearings • Use markdown headers to divide content • Section 1: Software and platform • The software you used (e.g. Python via Google Colab or VS Code) • Any packages that need to be installed (e.g. pandas, numpy, scikit-learn, matplotlib, seaborn) • The platform you used (Windows, Mac, or Linux) • Section 2: Map of your documentation - a folder tree listing all files and subfolders • Section 3: Step-by-step instructions to reproduce your results, written in plain English

SCRIPTS folder	• Goal: Contain all source code for the project • Include all notebooks used; name them in the order they should be run (e.g. 1_EDA.ipynb, 2_Model.ipynb) • Each script should include a header comment explaining its purpose • Include comments throughout explaining what each step does
DATA folder	• Goal: Contain all data for the project • Include the original dataset (CA_Weather_Fire_Dataset_1984-2025.csv) and the cleaned dataset (cleaned_df.csv) • Include a Data Appendix as a PDF with: • A statement of the unit of observation (what each row represents) • A subsection for each variable describing what it measures, its data type, and summary statistics
OUTPUT folder	• Goal: Contain all output generated by your analysis • Include all figures produced during EDA and model evaluation • Include confusion matrices, ROC curve, and feature importance plot • Use informative file names (e.g. logistic_confusion_matrix.png, roc_curve.png)
Written Report	• Include a PDF report in the OUTPUT folder with the following sections: • Introduction: problem context and motivation (half page maximum) • Data & EDA: summary of the dataset and key findings from exploration • Methods: description of how each model was built and trained • Results: evaluation metrics and figures for both models • Recommendation: which model to deploy and why, with at least one acknowledged limitation • References: sources cited in IEEE format • Total length: 2-4 pages including figures and references
References	• All references should be listed at the end of the written report • Use IEEE documentation style • Cite all provided supplemental materials plus at least one source you found independently

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